

Week 3 Lab Worksheet

The objective of this Week 3 laboratory session was to explore wireless network emulation and transport-layer traffic analysis. By transitioning from standard Mininet to **Mininet-WiFi**, we evaluated the architectural differences between wired switches and wireless Access Points/Stations (**LO1**). Utilizing the Mininet-WiFi Python API, we programmatically instantiated a wireless topology incorporating mobility models (**LO2**). Furthermore, by leveraging iperf to generate controlled TCP and UDP workloads, we utilized **Wireshark** to critically capture and contrast the underlying packet-level behaviours, reliability mechanisms, and connection patterns of the transport layer (**LO3 & LO4**).

Learning Objectives

1. Evaluate and differentiate wired and wireless network emulation paradigms.
2. Programmatically instantiate wireless network topologies and mobility models using Python APIs.
3. Simulate and generate transport-layer application traffic over emulated wireless media.
4. Critically analyse and contrast protocol-level behaviours (TCP vs. UDP) using packet analysers.

1. Introduction to Mininet-WiFi (Wireless Experiments)

1. Open a terminal by clicking Activities and typing terminal and type **sudo mn --wifi** and enter the password 123 to start Mininet-wifi

```
student@7com1076:~/7COM1076/LAB1$ sudo mn --wifi
[sudo] password for student:
*** Adding stations:
sta1 sta2
*** Adding access points:
ap1
*** Configuring nodes...
*** Creating network
*** Adding controller
*** Adding hosts:

*** Adding switches:

*** Adding links:
(sta1, ap1) (sta2, ap1)
*** Starting controller(s)
c0
*** Starting L2 nodes
ap1 ...
*** Starting CLI:
mininet-wifi> █
```

2. Observe the difference between wired hosts and switches vs wireless stations and access points. Ping between two stations
 - a. `sta1 ping sta2`
Press CTRL + C to end transmission
 - b. Exit Mininet-wifi by typing `exit`

2. Introduction to Mininet-wifi Python API

1. Navigate to the 7COM2008/LAB2 and open a terminal by **right click - Open in Terminal**
 - a. `sudo python3 lab2.py`

You will see a Mininet-wifi plot opened with two stations (sta1,sta2), two servers (TCPS, UDPS), and two access points (ap1, ap2)

3. Application Traffic (TCP and UDP)

1. Open xterm for TCPS and sta2 using
 - a. `xterm TCPS sta2`

On TCPS xterm, run the following command to initiate a TCP server:

- b. `iperf -s -p 5566 -i 1`

On sta2, initiate a connection to the TCP server:

- c. `iperf -c 10.0.0.4 -p 5566 -t 10`

2. Observe the TCP traffic by opening Wireshark. Can you spot the TCP connection patterns? Refer to notes in Week 2 Lab worksheet Task 3.c.ii on how to open Wireshark and identify the correct port to capture.

3. Open xterm for UDPS and sta1 using
 - a. `iperf -s -u -p 7788 -i 1`

On sta1, initiate a UDP connection:

- b. `iperf -c 10.0.0.5 -u -p 7788 -t 10`

4. Observe the UDP traffic by opening Wireshark. Can you spot the UDP connection patterns? Refer to notes in Week 2 Lab worksheet Task 3.c.ii on how to open Wireshark and identify the correct port to capture

4. Extension Task (Important)

1. Read through and understand the Python script in LAB2.
 - a. Can you spot how access points have been created?
 - b. Can you spot how access points have been started?
 - c. Can you spot how stations are created?
 - d. How mobility scenario is created?
 - e. Make notes for above